VISUAL ANALYTICS: NETWORK ANALYSIS

AB MOSCA (THEY/THEM)

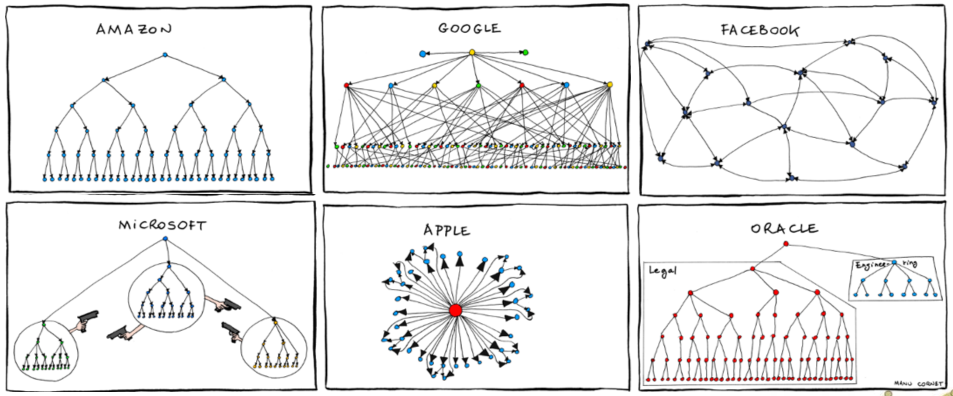
TODAY'S PLAN

- Mathematical intuition
- Modeling with graphs
- Visual analysis of networks

DEFINITION: GRAPH

In the mathematical sense a graph is a model for representing items and the relationships between those items. Ex:

- Social / friendship networks
- Computer networks
- Energy or transportation grids
- Organizational structures



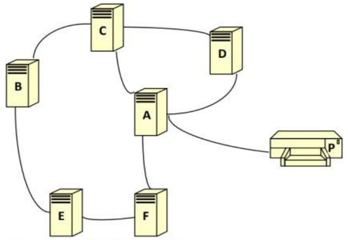
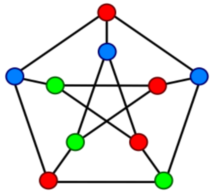
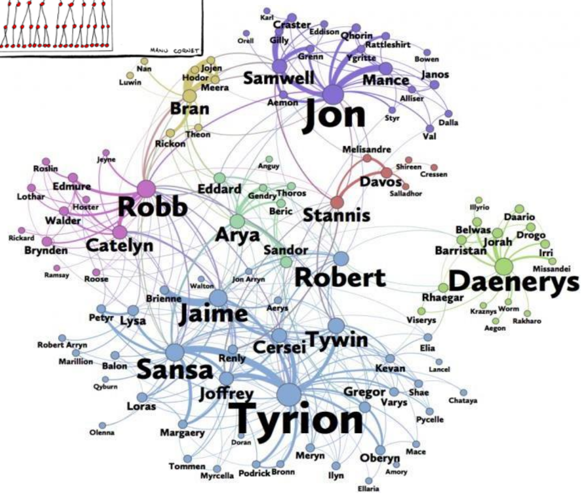


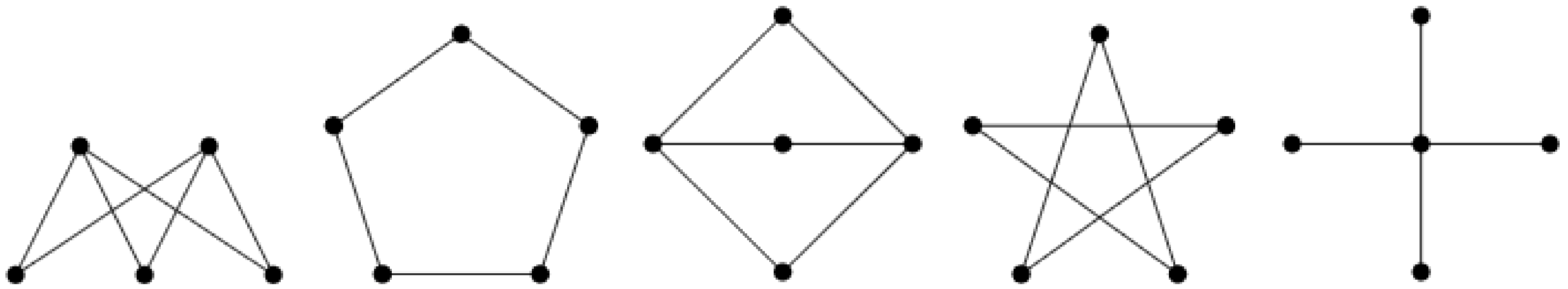
Figure 1- A simple computer network





MATHEMATICAL INTUITION

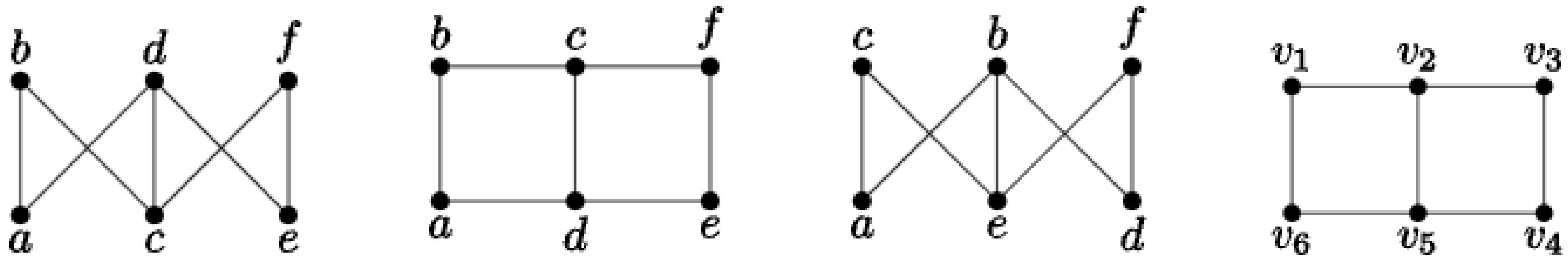
Are any of the graphs below the same? If you said yes, which ones and why?



MATHEMATICAL INTUITION

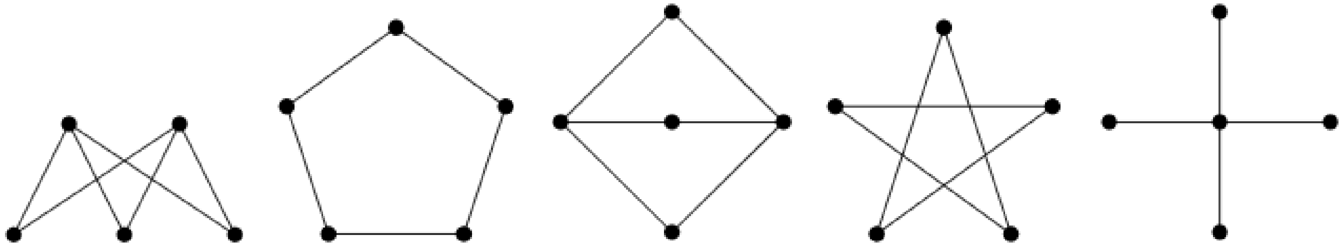
How about these with labels?

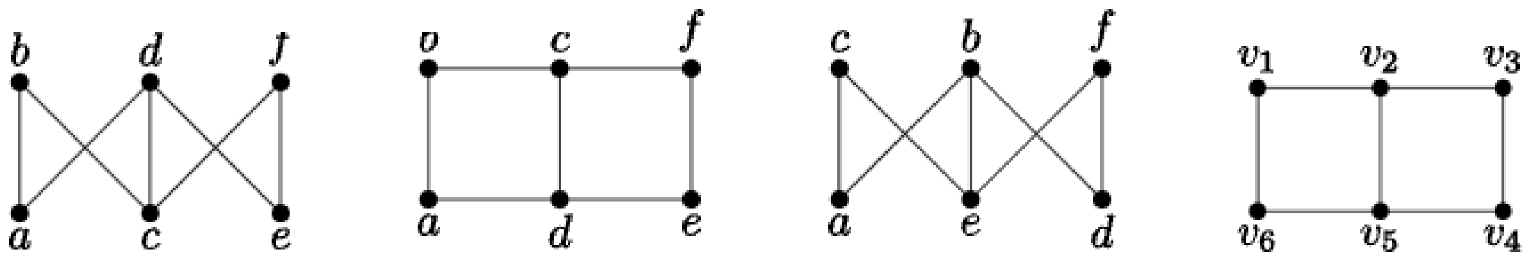
Are any of the graphs below the same? If you said yes, which ones and why?



MATHEMATICAL INTUITION

These examples are ***drawings*** of graphs.





Mathematically, a ***graph*** is an ordered pair, $G = (V, E)$, consisting of a nonempty set, V (called ***vertices***), and a set E (called ***edges***) of two-elements subsets of V.

(Vertices are also called nodes.)

MATHEMATICAL INTUITION

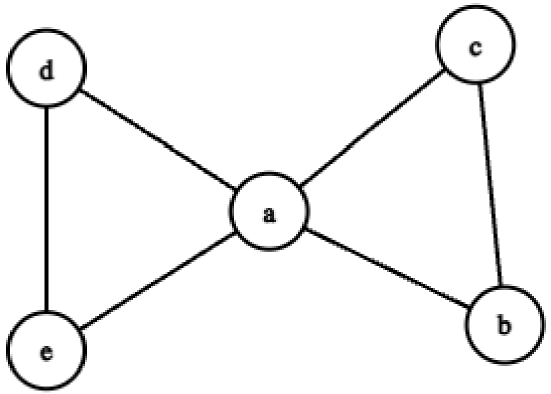
Mathematically, a ***graph*** is an ordered pair, $G = (V, E)$, consisting of a nonempty set, V (called ***vertices***), and a set E (called ***edges***) of two-elements subsets of V.

Draw this graph:
$V = \{v_1, v_2, v_3, v_4, v_5\}$
$E = \{\{v_1, v_3\}, \{v_1, v_5\}, \{v_2, v_4\}, \{v_2, v_5\}, \{v_3, v_5\}, \{v_4, v_5\}\}$

Ex. $V = \{a, b, c, d, e\}$,

$E = \{\{a, b\}, \{a, c\}, \{a, d\}, \{a, e\}, \{b, c\}, \{d, e\}\}$

To visualize this graph we might draw it:



MATHEMATICAL INTUITION

Mathematically, a ***graph*** is an ordered pair, $G = (V, E)$, consisting of a nonempty set, V (called ***vertices***), and a set E (called ***edges***) of two-elements subsets of V.

We can also represent a graph using an adjacency matrix:

$$\begin{array}{c c} & \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

What is the adjacency matrix for this graph?

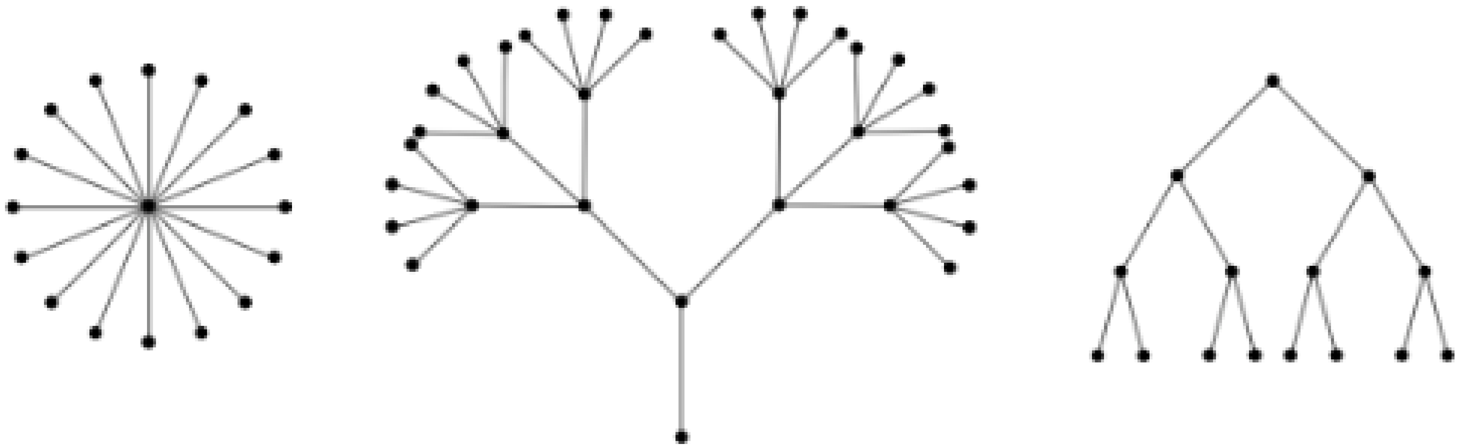
$V = \{v_1, v_2, v_3, v_4, v_5\}$

$E = \{\{v_1, v_3\}, \{v_1, v_5\}, \{v_2, v_4\}, \{v_2, v_5\}, \{v_3, v_5\}, \{v_4, v_5\}\}$

MATHEMATICAL INTUITION

A ***tree*** is a connected graph containing no cycles.

A ***forest*** is a graph containing no cycles.



https://ptwiddle.github.io/Graph-Theory-Notes/s_intro_trees.html

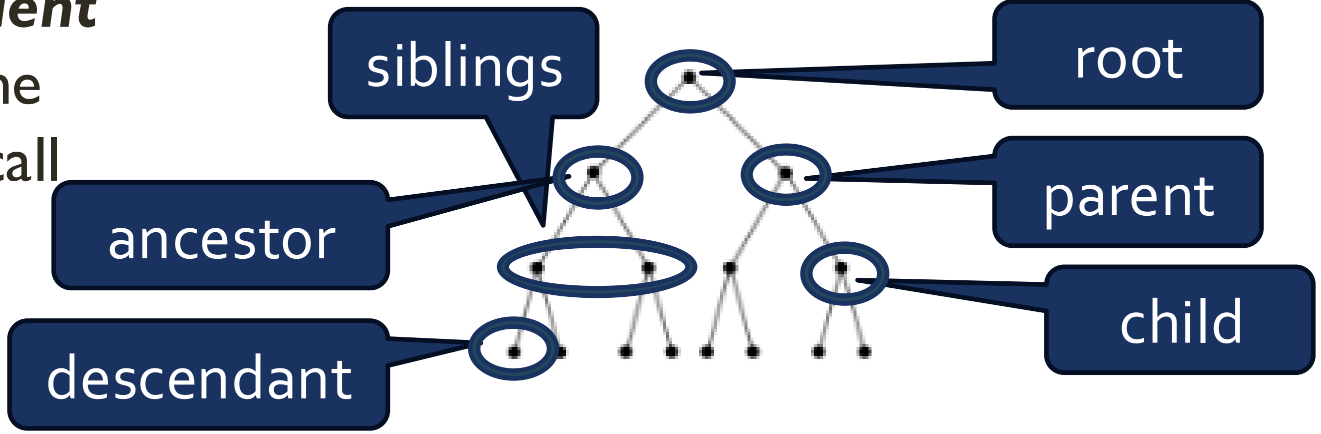
MATHEMATICAL INTUITION

We identify one vertex in a tree as the ***root***. Then, every other vertex on the tree can be characterized by its position relative to the root.

If two vertices are adjacent, we say the one closer to the root is the ***parent***, and the other is the ***child***.

Vertices with the same parent are ***siblings***.

In general, we say a vertex, v, is a ***descendent*** of a vertex, u, provided u is a vertex on the path from v to the root. Then, we would call u an ***ancestor*** of v.



RECALL -- DEFINITION: GRAPH

In the mathematical sense a graph is a model for representing items and the relationships between those items. Ex:

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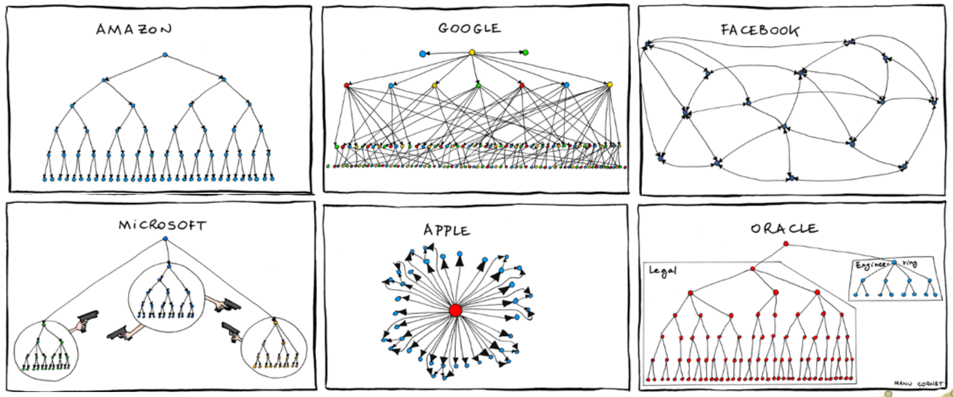


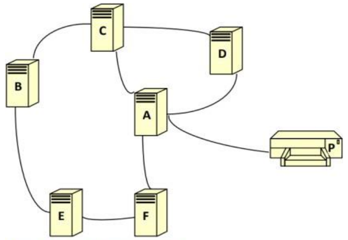
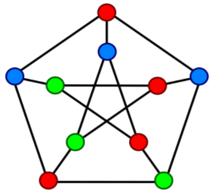
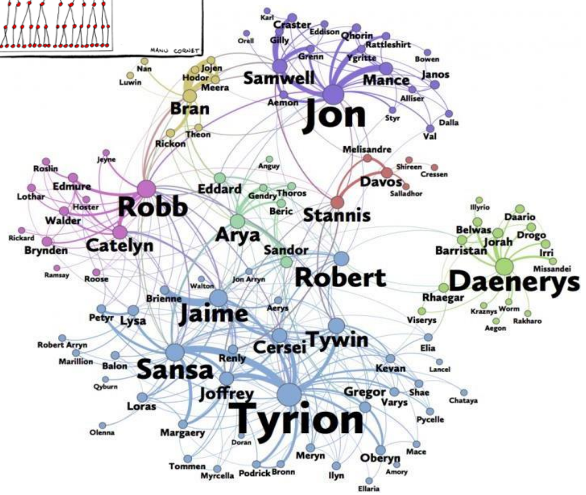


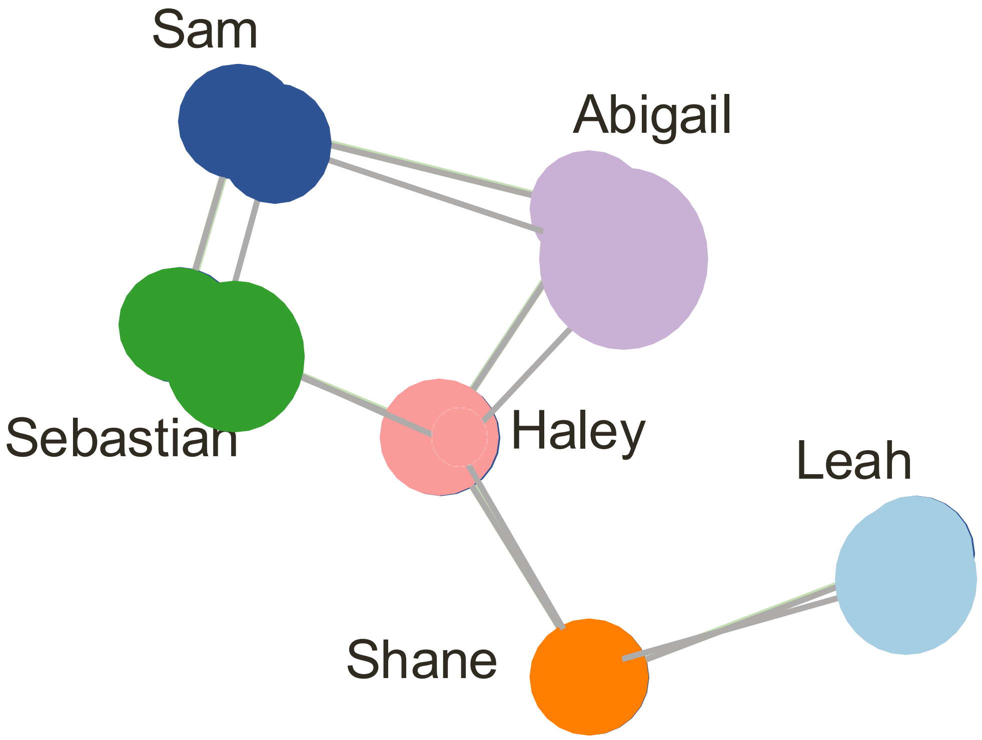
Figure 1- A simple computer network





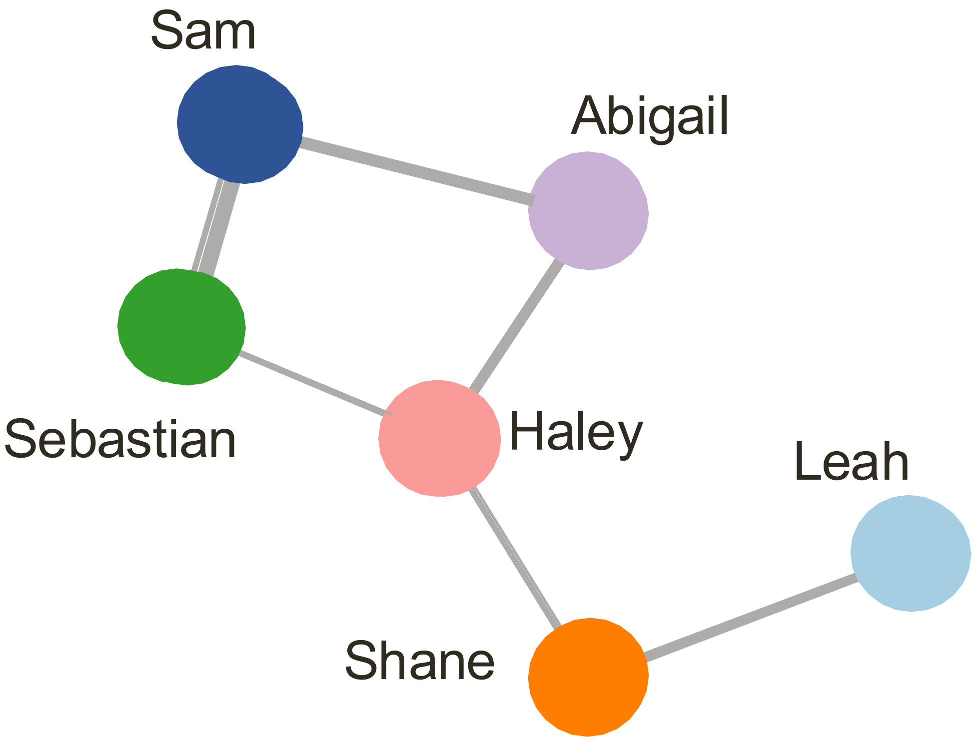
MODELING WITH GRAPHS

- **Node data:** There may be many attributes associated with each node



MODELING WITH GRAPHS

- **Edge data:** There may be many attributes associated with each edge

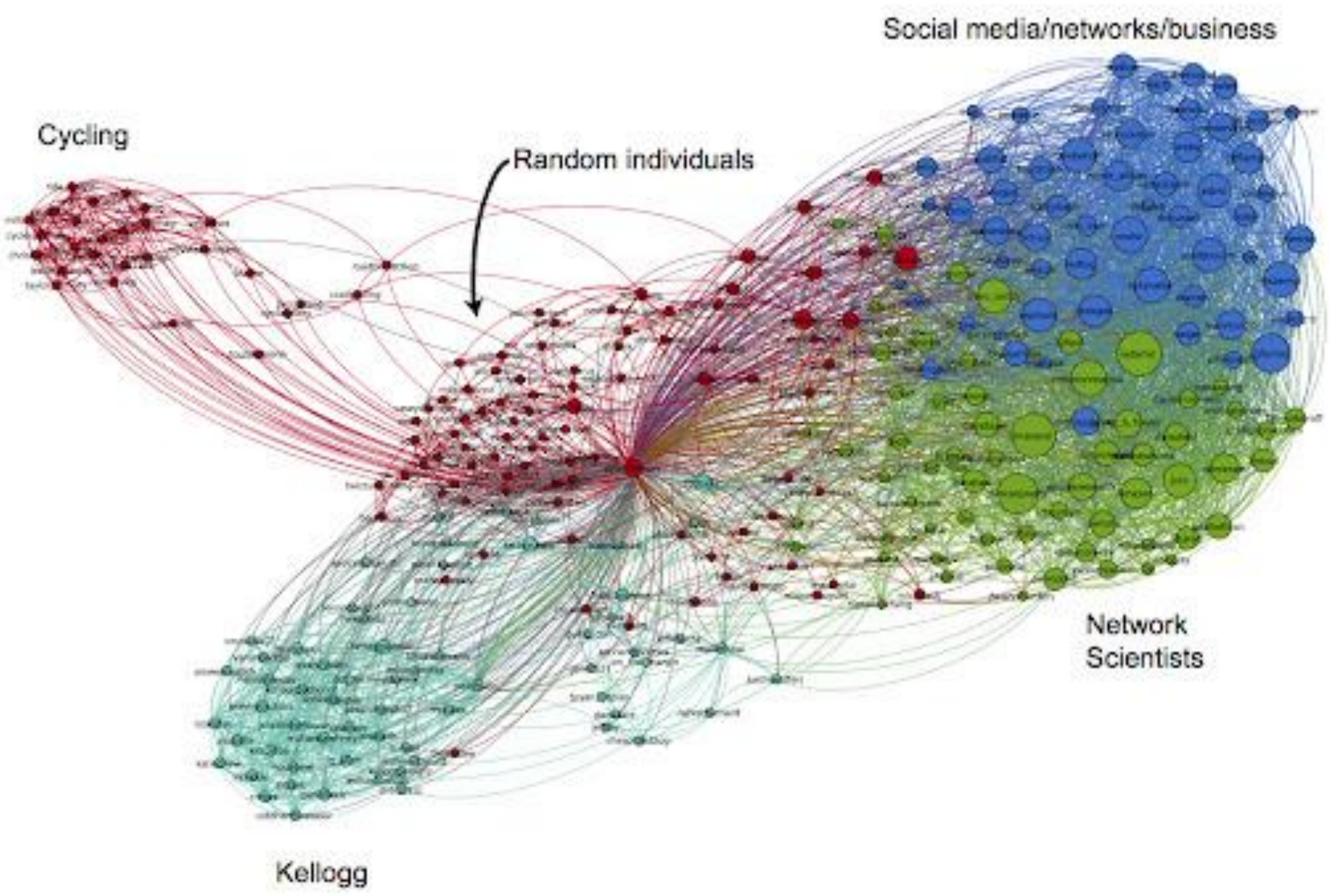


VISUAL ANALYSIS

What might visual analysis of networks reveal?

VISUAL ANALYSIS: INSIGHTS

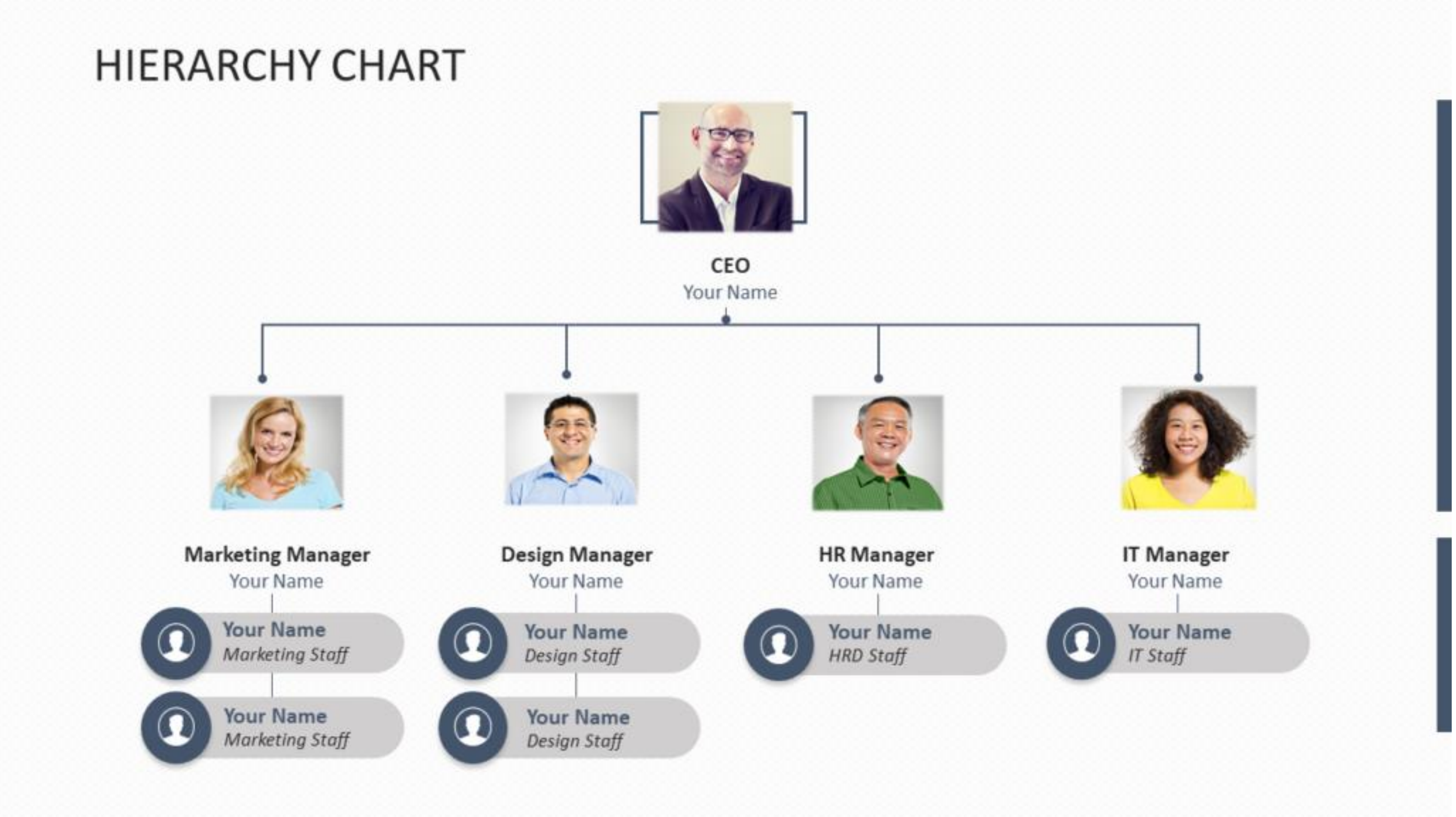
- **Highlight communities**



https://www.google.com/url?sa=i&url=http%3A%2F%2Fsocial-dynamics.org%2Ftag%2Fnetwork-visualization%2F&psig=AOvVaw1csnV4heoey3cU3srQTMiE&ust=1724861687547000&source=images&cd=vfe&opi=89978449&ved=0CBQQjRxqFwoTCOij7NPIlYgDFQAAAAAdAAAAABA5

VISUAL ANALYSIS: INSIGHTS

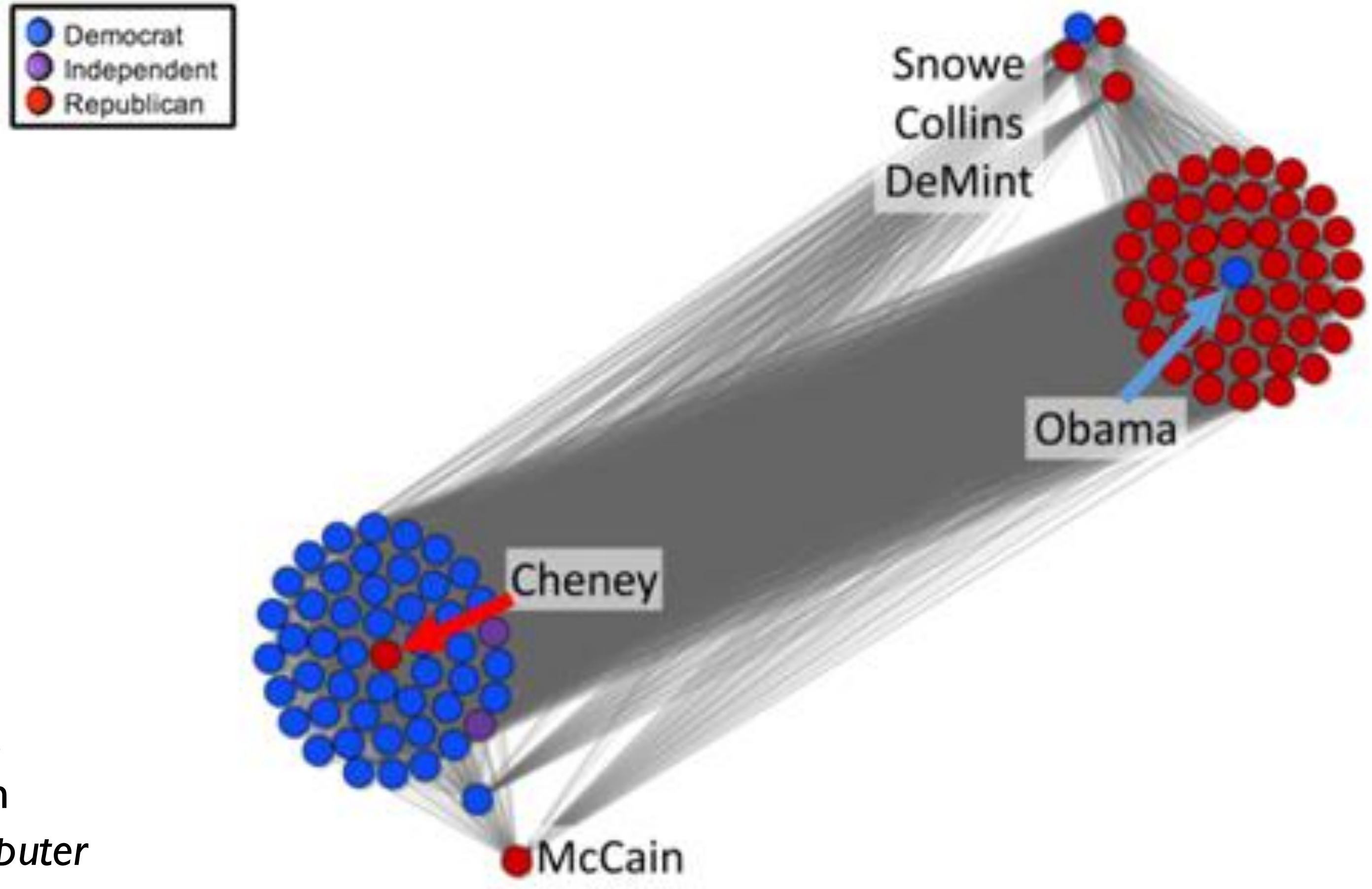
- **Illustrate hierarchy**



https://www.google.com/url?sa=i&url=https%3A%2F%2Fslideuplifts.medium.com%2Fmust-have-org-charts-to-visualize-your-project-team-structure-d61df5eb00f5&psig=AOvVaw2MFNBDY9jaG6RGpyqTDuKD&ust=1724861970547000&source=images&cd=vfe&opi=89978449&ved=0CBgQ3YkBahcKEwiYpfLayZWIAxUAAAAAHQAAAAAQCQ

VISUAL ANALYSIS: INSIGHTS

- **Reveal outliers**



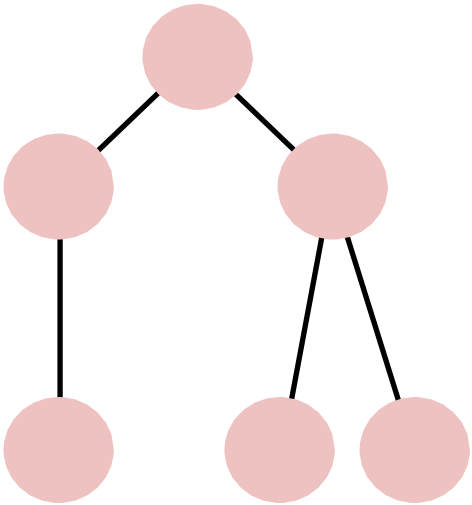
A. Suh, M. Hajij, B. Wang, C. Scheidegger and P. Rosen, "Persistent Homology Guided Force-Directed Graph Layouts," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 26, no. 1, pp. 697-707, Jan. 2020, doi: 10.1109/TVCG.2019.2934802.

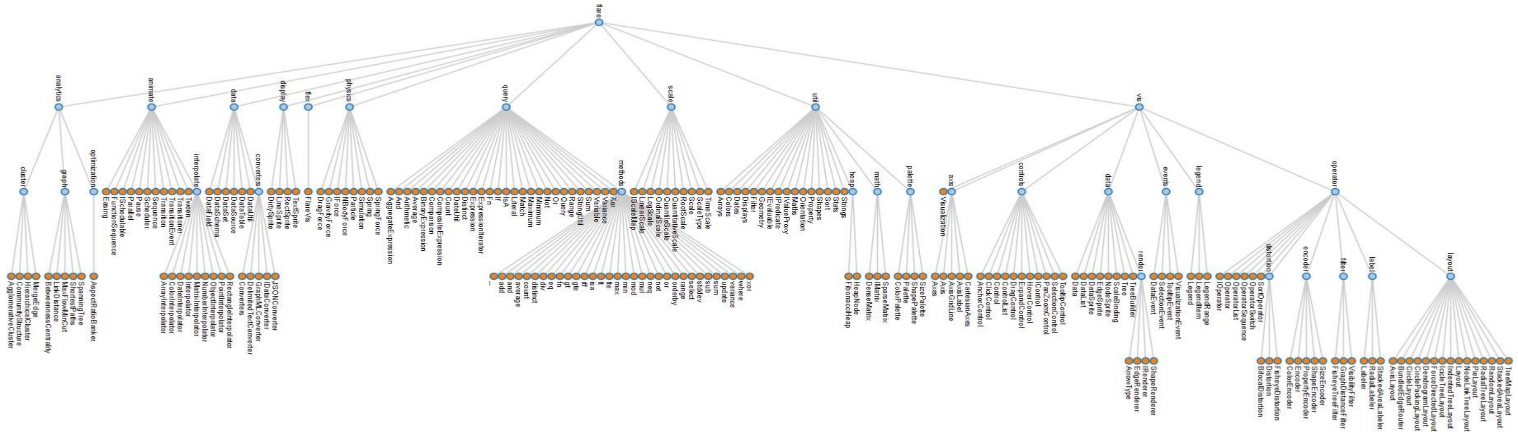
VISUAL ANALYSIS: TECHNIQUES

- There are many ways to encode and layout graph visualizations.
- The correct one depends on what tasks you need to support or what features of the data need to be highlighted.

VISUAL ANALYSIS: TECHNIQUES

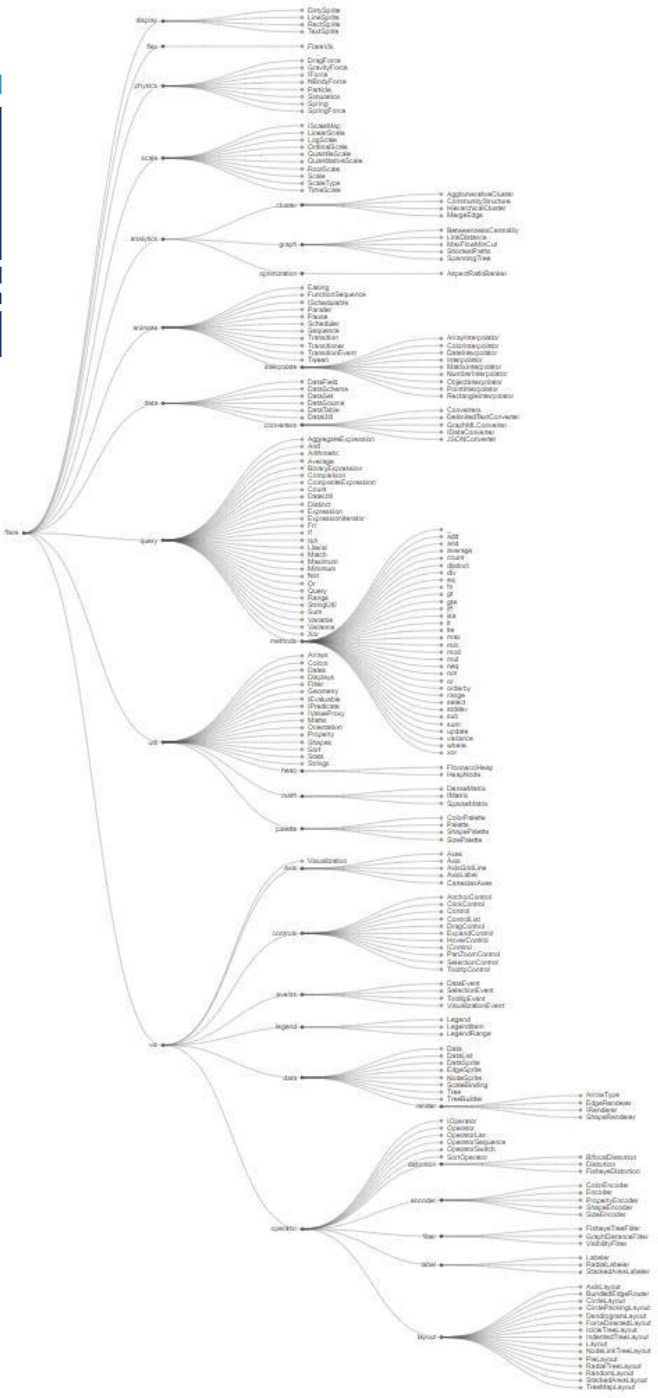
- **Node-link tree diagrams**
  - Nodes are distributed in space, connected by straight or curved lines
  - Typical approach is to use 2D space to break apart breadth and depth
  - Often, position is used to communicate hierarchical orientation

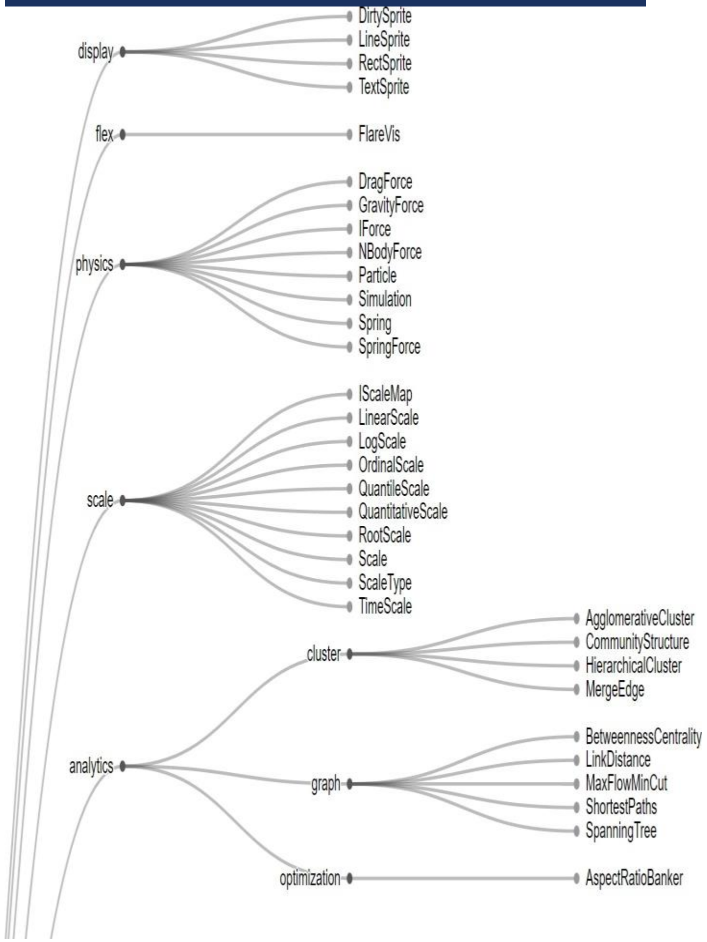




VISUAL ANALYSIS: TECHNIQUES

- **Node-link tree diagrams**
  - Algorithms, such as the Reingold-Tilford algorithm, efficiently calculate optimal layout



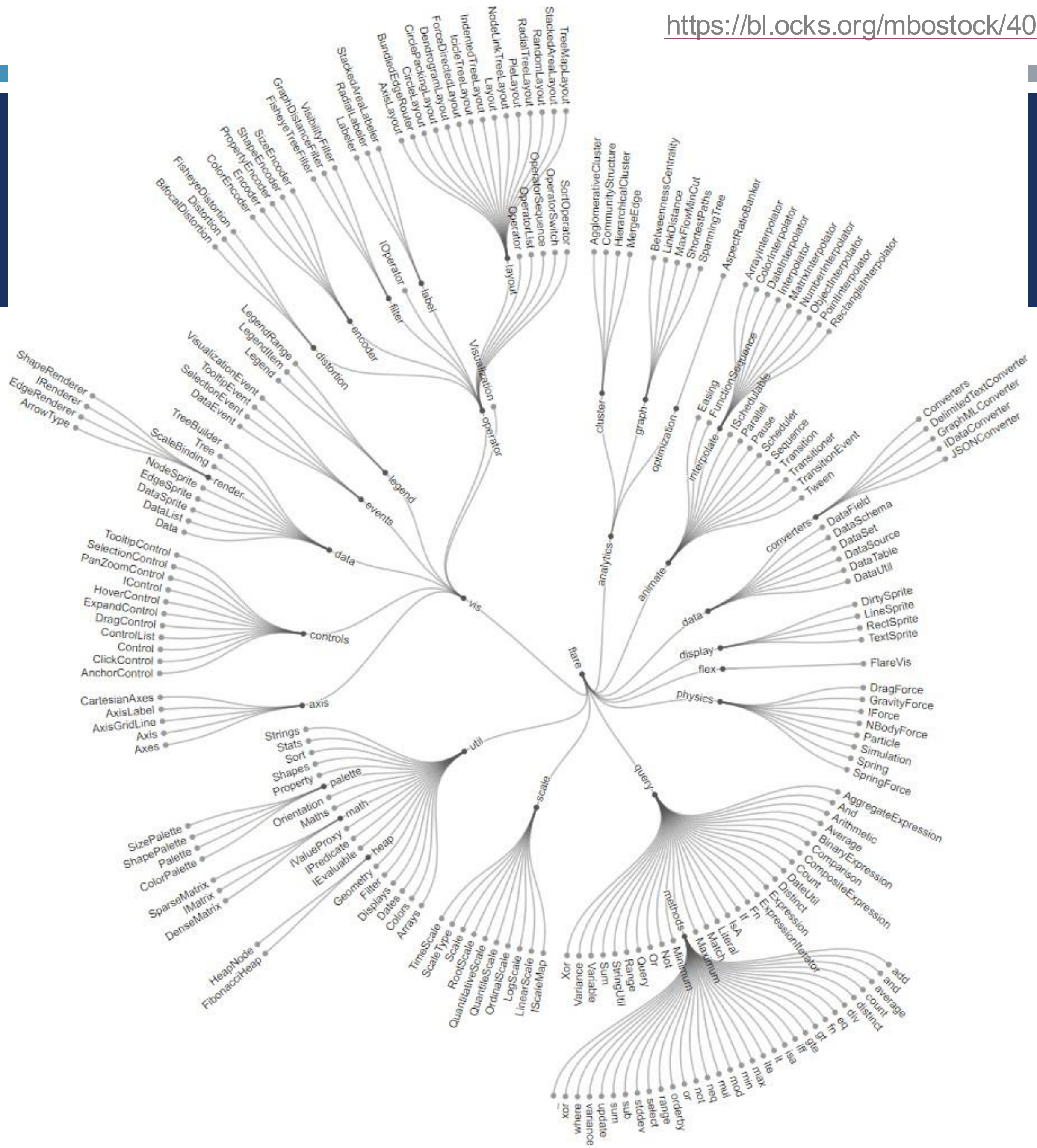


https://bl.ocks.org/mbostock/4339184

https://bl.ocks.org/mbostock/4063550

VISUAL ANALYSIS: TECHNIQUES

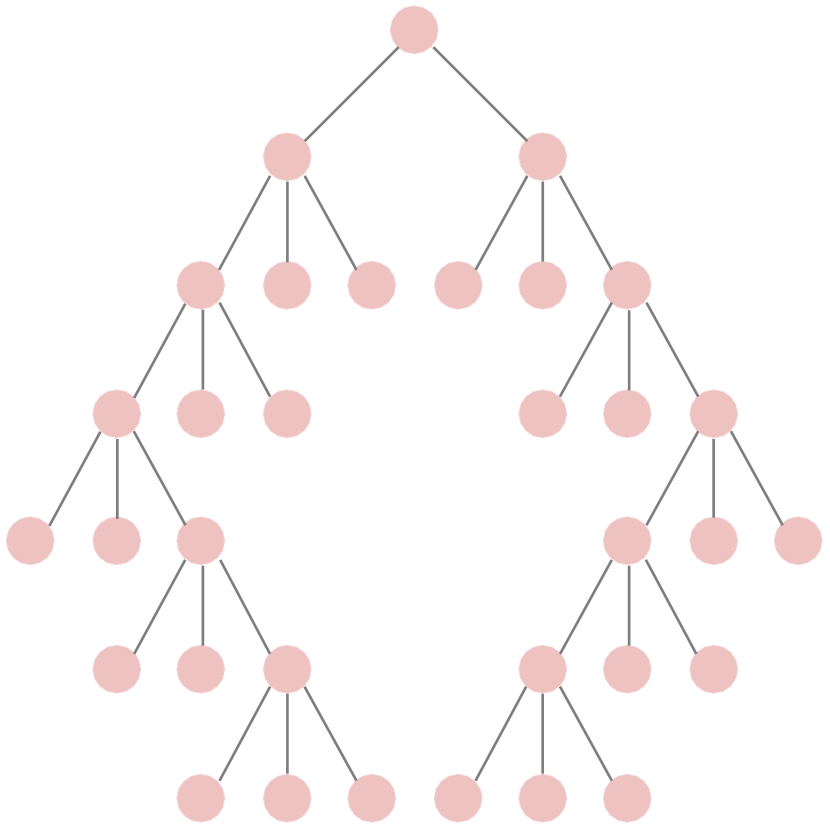
- **Node-link tree diagrams**
  - Layout can also be radial



https://bl.ocks.org/mbostock/4063550

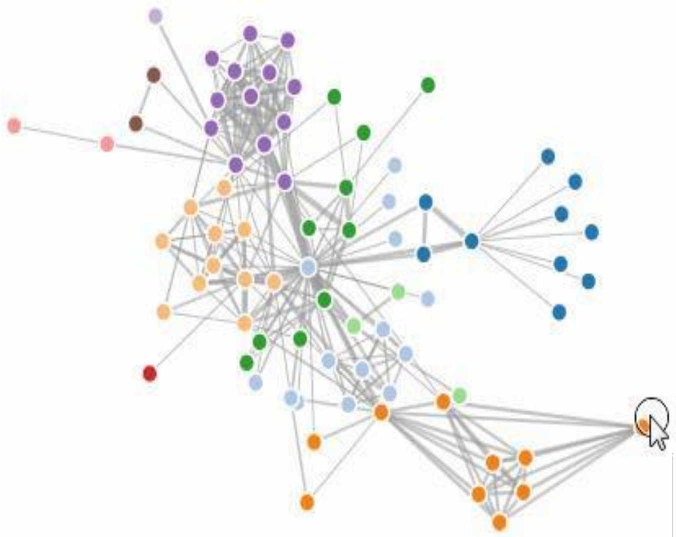
VISUAL ANALYSIS: TECHNIQUES

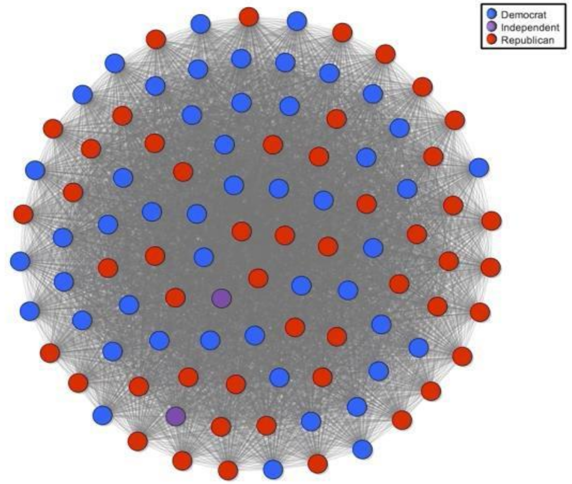
- **Node-link tree diagrams**
  - Reingold-Tilford algorithm
    - Bottom-up recursive approach
      - Repeatedly divide space by leaf count
    - For each parent, make sure subtrees are drawn
    - Make smarter use of space
      - Maximize density and symmetry
      - Clearly encode depth level
      - No edge crossings
      - Pack subtrees as closely as possible
      - Centers parent over subtrees



VISUAL ANALYSIS: TECHNIQUES

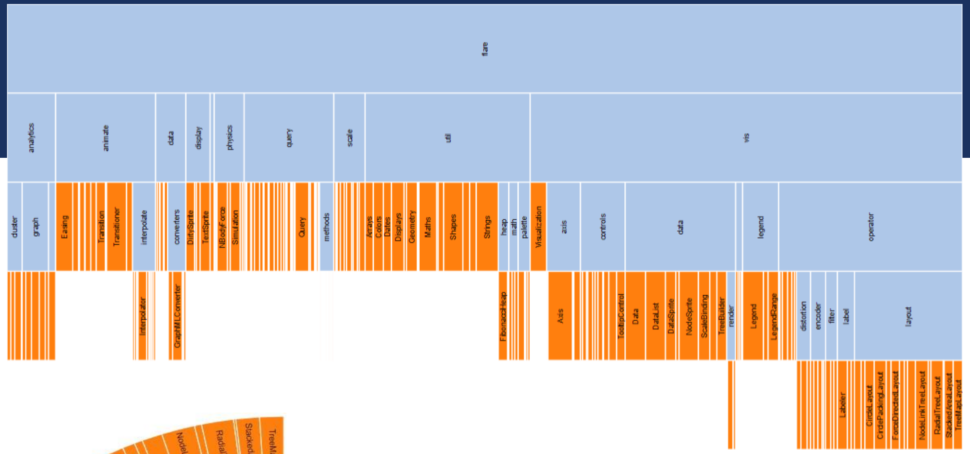
- **Node-link tree diagrams**
  - Force-directed layouts model layout with attraction (via edges) and repulsion (between nodes) to optimize layout
  - However, many edges can cause visual clutter

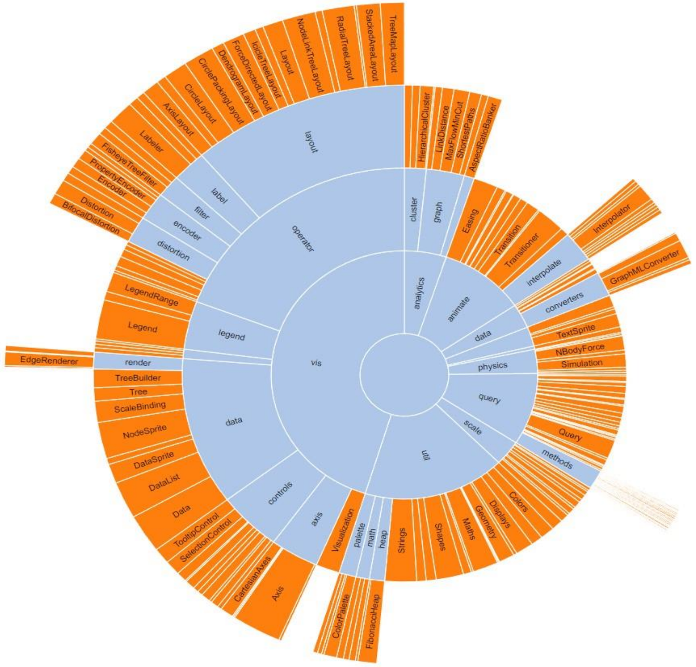




VISUAL ANALYSIS: TECHNIQUES

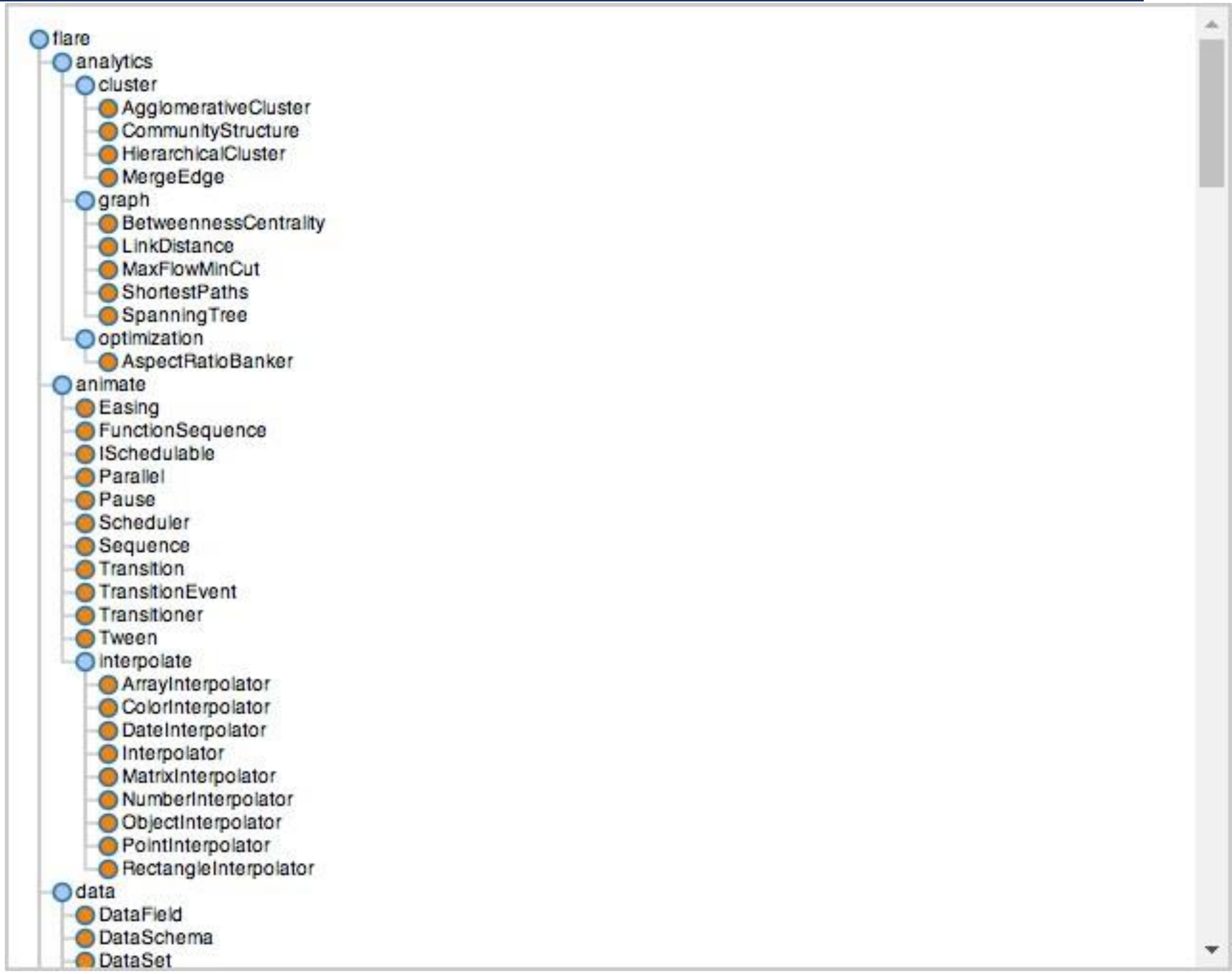
- **Layered (adjacency) Diagrams**
  - Space-filling variant of node-link diagrams
  - Nodes drawn as solid areas (arcs or bars)
  - Placement relative to adjacent nodes reveals place in hierarchy
  - Root node at top / center
  - Leaf nodes at bottom





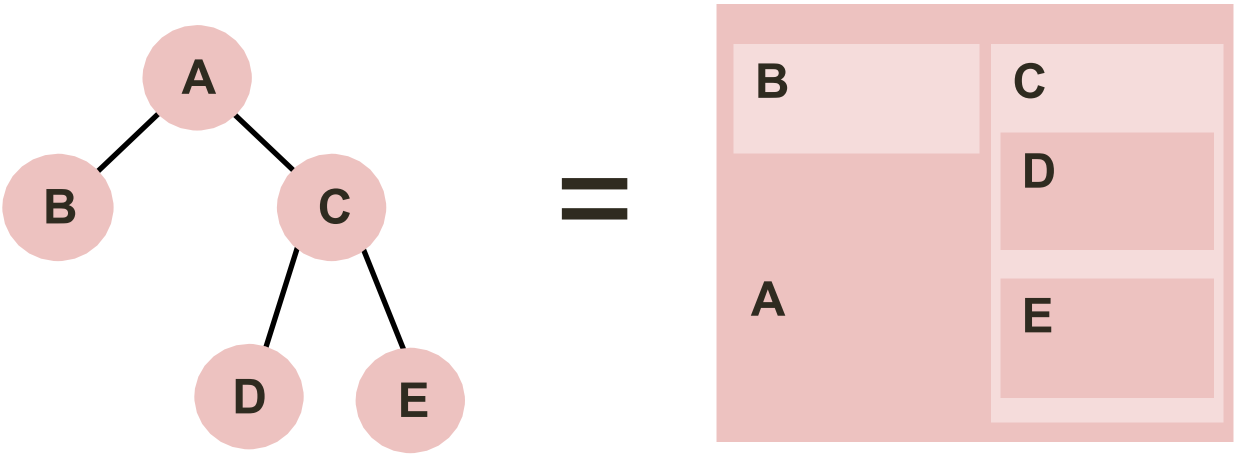
VISUAL ANALYSIS: TECHNIQUES

- **Indented List**
  - Highlights parent / child relationships
  - Simple to compute
  - Potentially a lot of scrolling



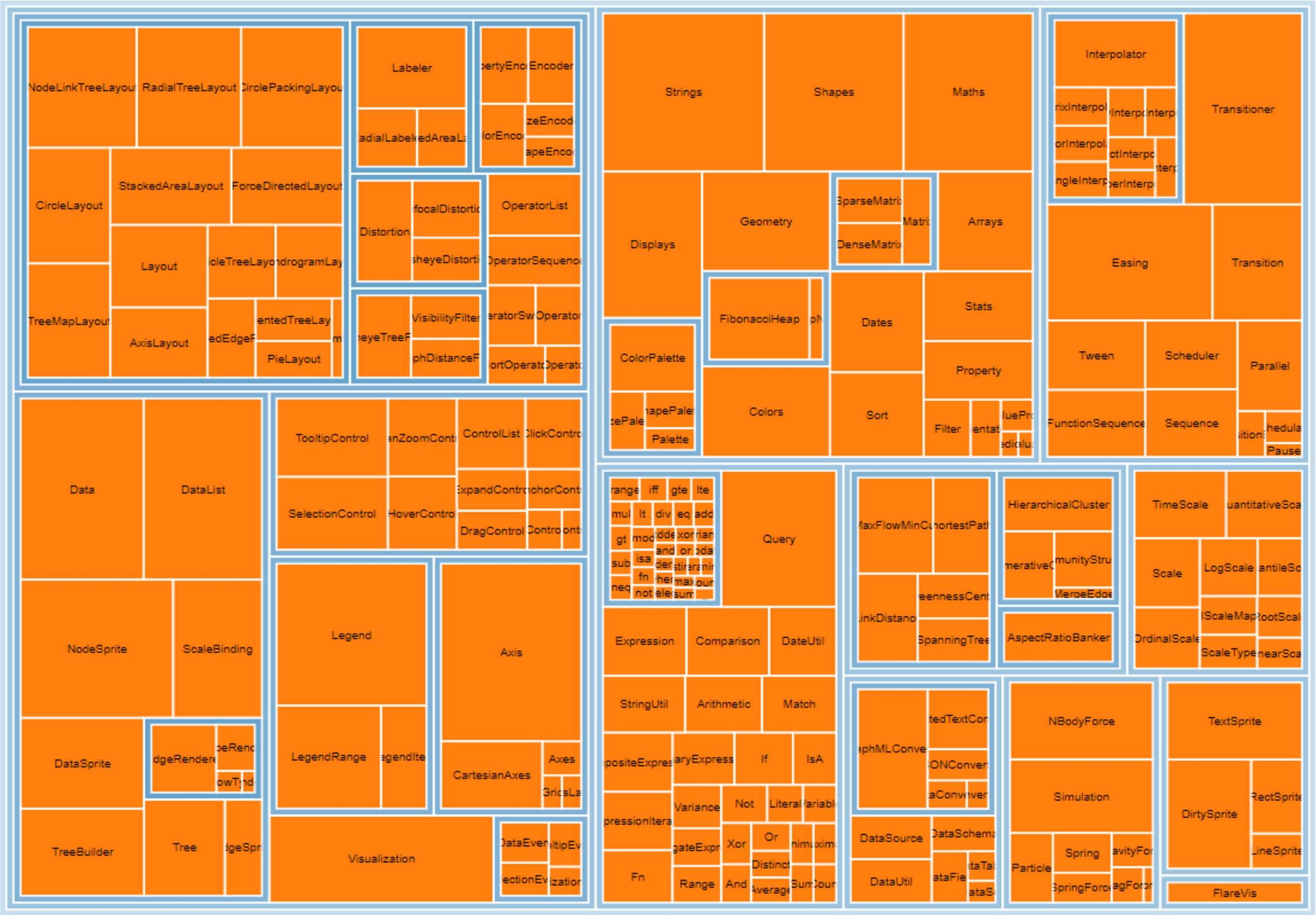
VISUAL ANALYSIS: TECHNIQUES

- **Enclosure (treemap) Diagram**
  - Encodes tree structure using spatial enclosure
    - Enclosure indicates hierarchy
  - Provides a single view of entire tree
  - Easier to spot small / large nodes (i.e. node with many or few descendants)



VISUAL ANALYSIS: TECHNIQUES

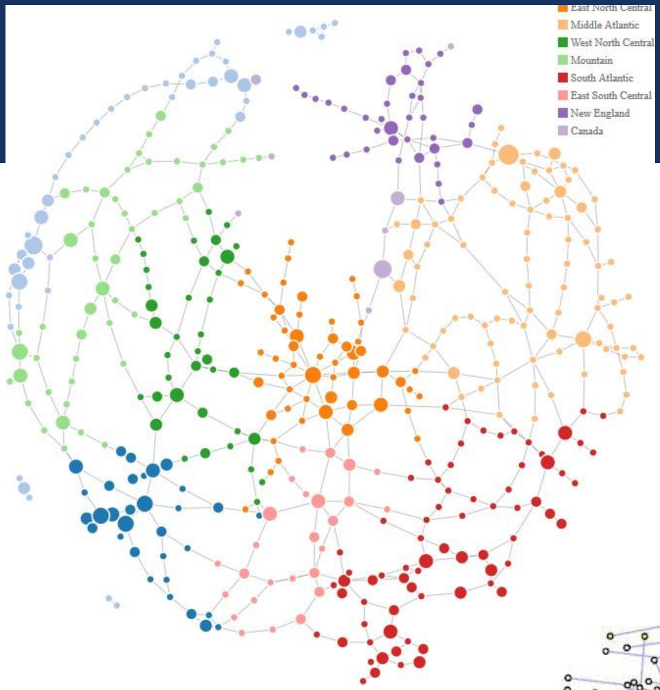
- **Enclosure (treemap) Diagram**
  - Can obscure depth



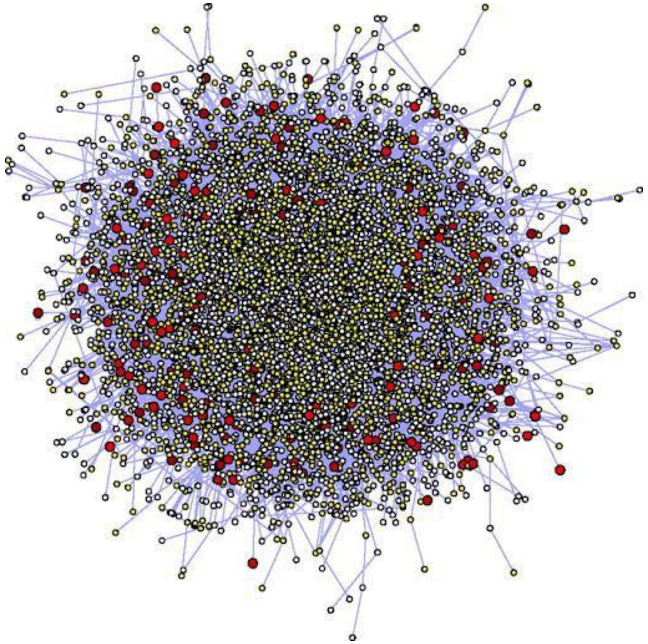
VISUAL ANALYSIS: TECHNIQUES

- Creating an effective visualization depends on the primary goal of analysis
- Each method of visual analysis has tradeoffs

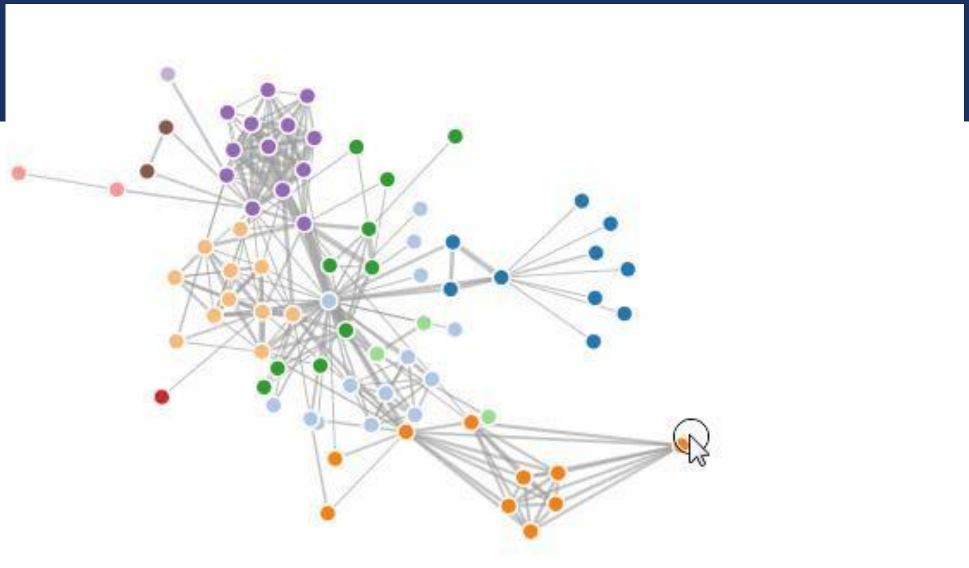
VISUAL ANALYSIS: TECHNIQUES

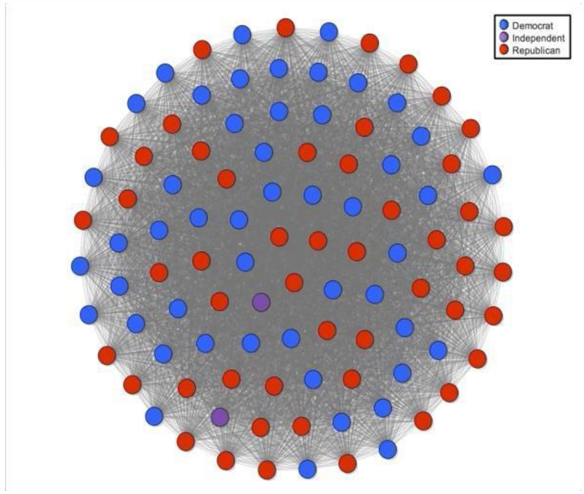


- Each method of visual analysis has tradeoffs
- **Reingold-Tilford algorithm**
  - **PROS**:
    - understandable visual mapping
    - shows overall structure, clusters, paths
    - flexible, many variations / layouts
  - **CONS**:
    - most trivial algorithms are slow
    - not good for dense (very connected) graphs



VISUAL ANALYSIS: TECHNIQUES

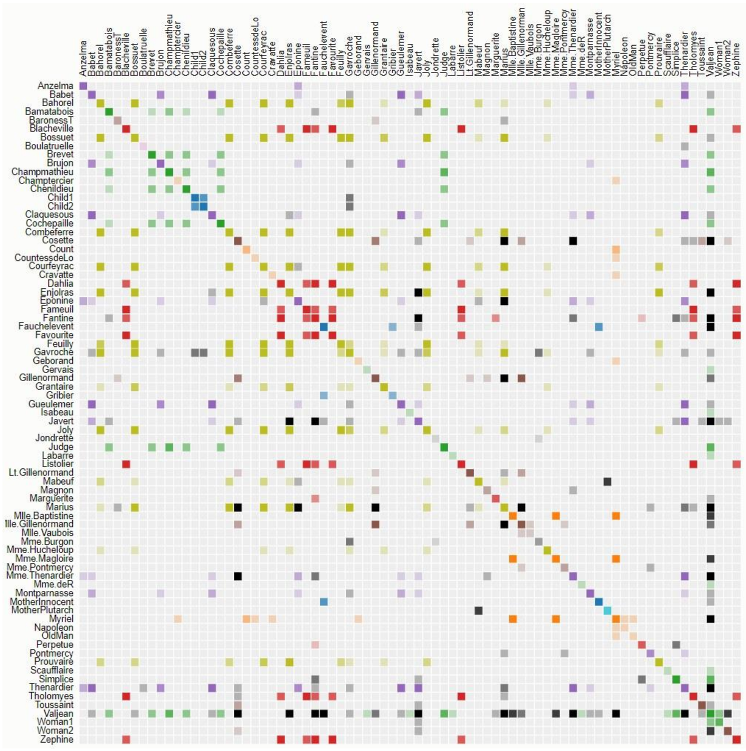


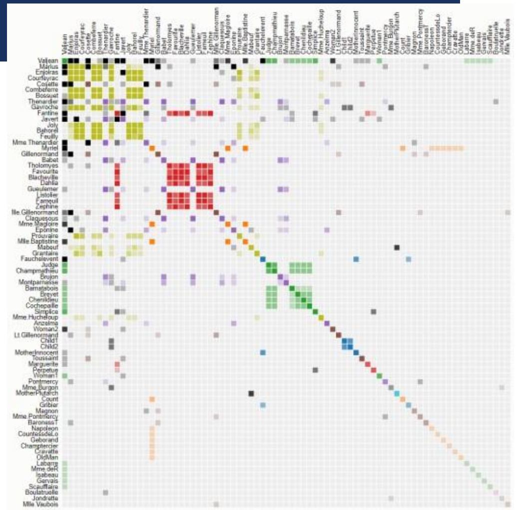


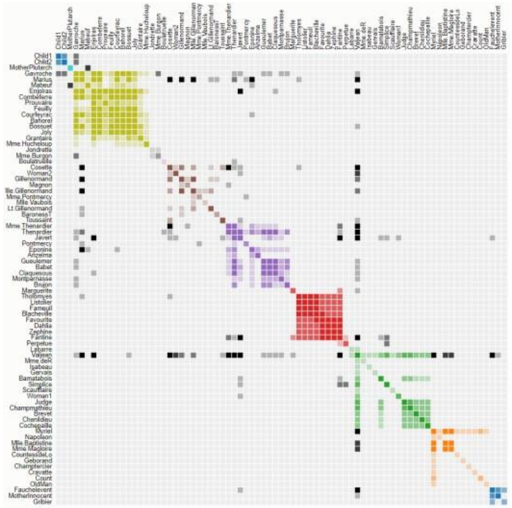
- Each method of visual analysis has tradeoffs
- **Force-directed Layout**
  - **PROS**:
    - aesthetically-pleasing
    - interactive (pull & drag!)
    - automatic & flexible
  - **CONS**:
    - forces are computationally expensive ~ O(n2)
    - doesn't work well on dense graphs

VISUAL ANALYSIS: TECHNIQUES

- Each method of visual analysis has tradeoffs
- **Adjacency Diagram**
  - **PROS**:
    - great for dense graphs
    - visually scalable
    - can spot clusters
  - **CONS**:
    - row order affects what you can see
    - abstract visualization
    - hard to follow paths







<u>Why does ordering matter?</u>
(above) layout ordered by Name
(left, top) layout ordered by Frequency
(left, bot) layout ordered by Cluster

ACTIVITY

- Your Turn:
  - Find a visualization of network data
  - Re-draw the visualization using a different method than the original
  - Does anything become apparent in the re-draw that was hidden in the original? Which do you think is more effective? Why?